

Product Requirements Document

AI-Powered Governance Intelligence Platform

Project Overview & Context

Government programmes across India involve multiple ministries, departments, implementing agencies, and geographic regions. Data related to schemes, budgets, projects, beneficiaries, implementation progress, and outcomes is fragmented across portals, reports, financial systems, and geospatial datasets.

This platform integrates data from multiple government schemes—anchored on canonical Local Government Directory (LGD) codes—to provide a unified view of implementation, resource utilization, geographic coverage, and outcomes.

Ingested Datasets (MVP Baseline)

- 1. **MGNREGA (Ministry of Rural Development):** Wage employment, transactional progress, person-days generated, and wage expenditure.
- 2. **PM-KISAN (Ministry of Agriculture):** Direct Benefit Transfer (DBT) income support releases, landholder verification counts.
- 3. **PMAY-G (Ministry of Rural Development):** Rural housing sanctioning, geo-tagged construction milestones, and tranche releases.

1. Personas & Permissions (Role-Based Access Control)

Persona Role	Target User	Data Access Scope	Permissions & Capabilities
Super Admin	Platform Engineering / NIC Admin	Global System-wide	Schema migrations, API token management, pipeline scheduling, audit log inspection, user provisioning.
Central Executive	Cabinet Secretariat, Ministry Secretaries, NITI Aayog	Pan-India, Cross-Ministry	Read-only aggregate and drill-down access across all schemes, NLP query workbench, automated convergence reports, cross-scheme correlation exports.

Persona Role	Target User	Data Access Scope	Permissions & Capabilities
Department Nodal Officer	Joint Secretary / Director (e.g., MoRD, MoA)	Ministry-Specific (Deep) + Cross-Ministry (Aggregated)	Full access to own ministry's granular data; read-only macro indicators for other ministries; set threshold alerts for scheme milestones.
District Magistrate (DM / DC)	District Collectors, Chief Development Officers (CDOs)	District-Specific (Down to Gram Panchayat)	Granular cross-scheme view within assigned district LGD code; field intervention tracking, flagging reporting discrepancies, assigning field audits.
Auditor / Research Analyst	CAG, Evaluation Offices, Independent Researchers	Anonymized Pan-India	Read-only access to analytical views, execution of NL-to-SQL queries with citation verification, export of anomaly audit trails. PII completely masked.

2. Feature Prioritization (MoSCoW Breakdown)

Must-Have (MVP)

- **Automated LGD Harmonization Engine:** Ingests CSV/JSON data from 3 schemes and resolves entity names against canonical LGD codes.
- **Unified Cross-Scheme Data Model:** Harmonized schema linking Ministries, Schemes, Budgets, Implementations, and Outcomes.
- **Executive Cross-Ministry Dashboard:** Interactive choropleth map (India → State → District), key performance metric cards (Fund Allocated, Released, Utilized, Beneficiary Count).
- **Scheme Overlap & Gap Matrix:** Automated identification of duplicate benefit profiles (e.g., high MGNREGA work demand overlapping with stalled PMAY-G construction in the same block).
- **Natural Language Query Interface (NL-to-SQL):** Conversational AI interface backed by schema-aware LLM agents, outputting verified SQL with source-data citations.
- **Granular PostgreSQL Row-Level Security (RLS):** Administrative boundary isolation based on authenticated JWT session claims.

Should-Have (Phase 2)

- **Underperformance Anomaly Detector:** Automated Z-score analysis flagging districts where fund release vs. physical asset completion deviates by > 2 standard deviations from state median.
- **Traceable Evidence Drawer:** Deep-dive modal showing exact underlying database rows and calculation formulas supporting an AI-generated insight.
- **PDF Executive Brief Generator:** Automated scheduled generation of monthly convergence briefs for Cabinet Secretariat.

Could-Have (Phase 3)

- **Predictive Resource Reallocation Simulator:** "What-if" scenario builder forecasting asset creation if underutilized funds from Scheme X are redirected to Scheme Y.
- **Multilingual Query Interface:** Prompting and dashboards localized in Hindi and major regional languages.

Won't-Have (Out of Scope for MVP)

- Direct write-back integration into PFMS (Public Financial Management System) to release funds.
- Public-facing citizen grievance ticketing portal.
- Live biometrics processing or raw Aadhaar data storage.

3. User Stories & Acceptance Criteria

US-01: Cross-Ministry Overlap & Convergence Analysis

- **As a** Central Executive (Cabinet Secretariat),
- **I want to** view an interactive overlap matrix of MGNREGA asset creation and PMAY-G rural housing completion by district,
- **So that** I can identify administrative blocks with high wage disbursement but lagging housing infrastructure.

Acceptance Criteria

- **Given** that the user has selected financial year 2025-2026 and filtered by the state of Odisha,
- **When** they navigate to the "Convergence & Overlap" view,
- **Then** the platform renders a bivariate choropleth map highlighting districts with >80% MGNREGA fund utilization alongside <40% PMAY-G milestone completion,
- **And** provides an exportable summary table listing the specific District LGD codes and responsible state nodal departments.

US-02: Traceable Natural Language Querying

- **As an** Evaluation Analyst,
- **I want to** type a plain-language question such as *"Show me districts where PM-KISAN cash transfer disbursement exceeds 90% but fertilizer subsidy intake dropped in Q3"*,
- **So that** I do not have to write custom SQL joins across heterogeneous departmental schemas.

Acceptance Criteria

- **Given** an authenticated session with Analyst permissions,
 - **When** the query is submitted in the NLP query input box,
 - **Then** the system outputs a synthesized data card with an interactive visualization,
 - **And** displays an inspectable "Evidence & Provenance" tab showing the generated SQL query, execution latency, and exact source tables (`pm_kisan_disbursements`, `fert_subsidy_records`) utilized.
-

US-03: Geographic Underperformance Anomaly Flagging

- **As a** District Magistrate,
- **I want to** receive automated alerts when any centrally sponsored scheme in my district stalls for more than 45 days,
- **So that** I can direct Block Development Officers (BDOs) to unblock administrative bottlenecks.

Acceptance Criteria

- **Given** that the DM is logged in with LGD claim `DISTRICT_2704` (e.g., Gorakhpur),
- **When** a background ingestion pipeline detects zero physical progress updates for PMAY-G house sanctions over a 45-day window,
- **Then** an anomaly badge is rendered on the DM's dashboard marked **CRITICAL : Stalled Milestone**,
- **And** clicking the badge displays the exact Gram Panchayats affected and the contact information of the assigned block engineer.